



LECTURE 9

PRODUCTION PLANNING SYSTEMS

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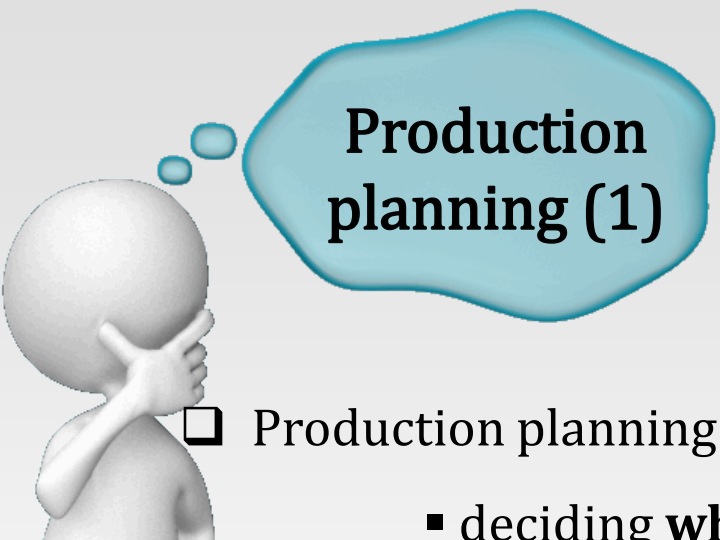
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Objectives

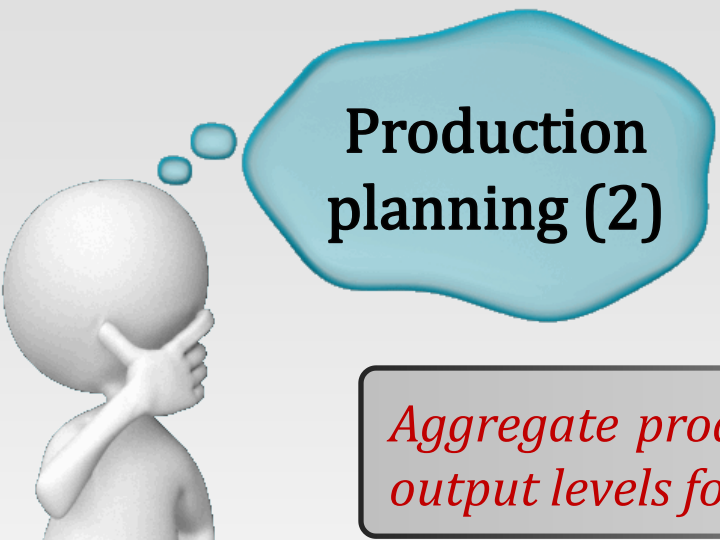
1. Knowing the production planning strategies.
2. Knowing the materials planning requirements.
3. Understanding the capacity adjustment methods used for short and long term.





Production planning (1)

- ❑ Production planning is concerned with:
 - deciding **which** products to make ,
 - **how many** of each,
 - **when** they should be completed,
 - **scheduling** the delivery and/or production of the parts and products,
 - **planning** the manpower and equipment **resources** needed to accomplish the production plan.
- ❑ Activities within the scope of production planning:
 - **aggregate** production planning
 - **master** production planning
 - **material requirements** planning
 - **capacity** planning



Production planning (2)

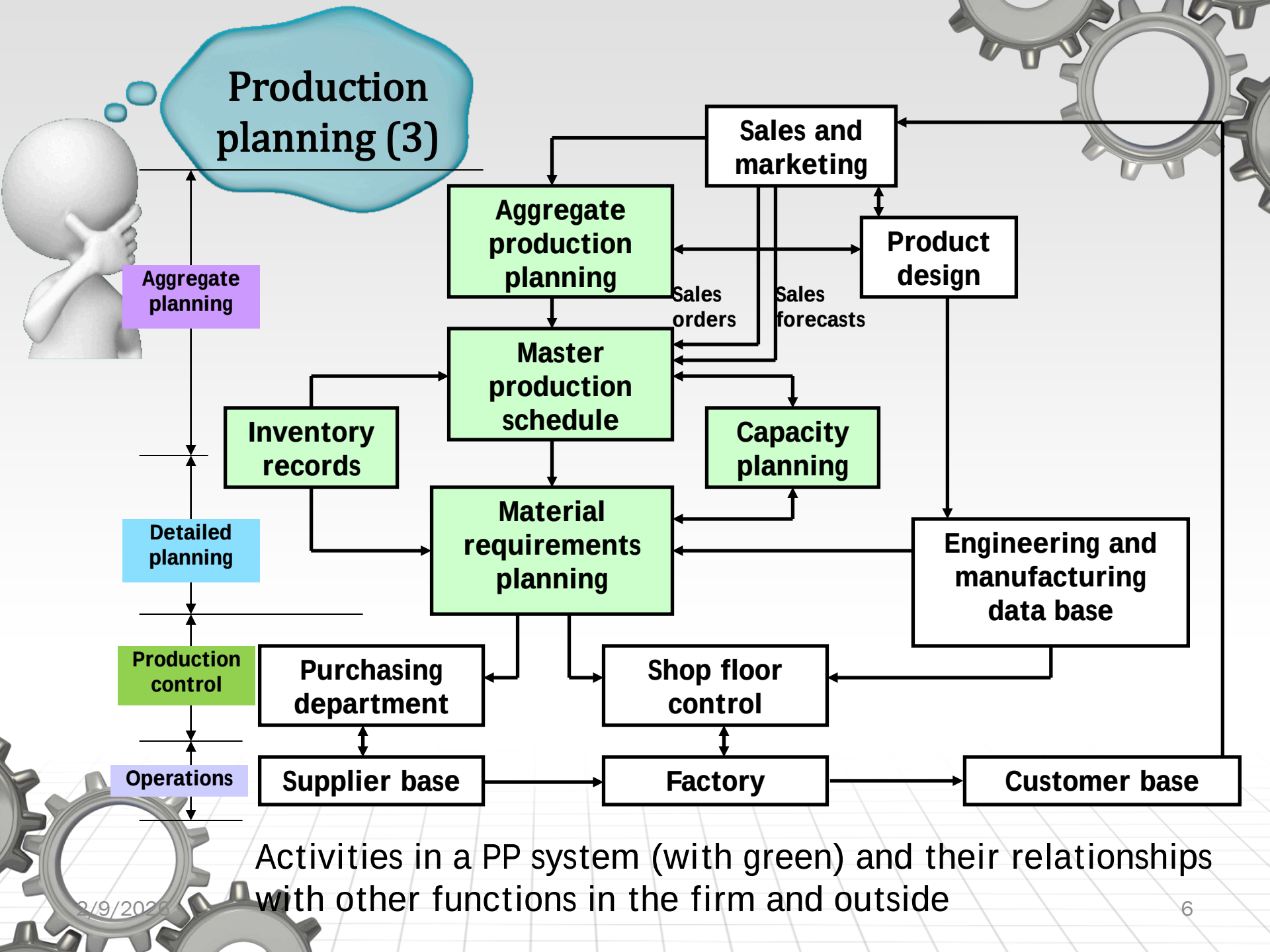
Aggregate production planning involves planning the production output levels for major product lines produced by the firm.



Master production planning is used to convert the aggregate production plan into a master production schedule (MPS) which is a specific plan of the quantities to be produced of individual models within each product line.

Material requirements planning (MRP) is a planning technique, usually implemented by computer that translate the MPS of end products into a detailed schedule for the raw materials and parts used in those end products.

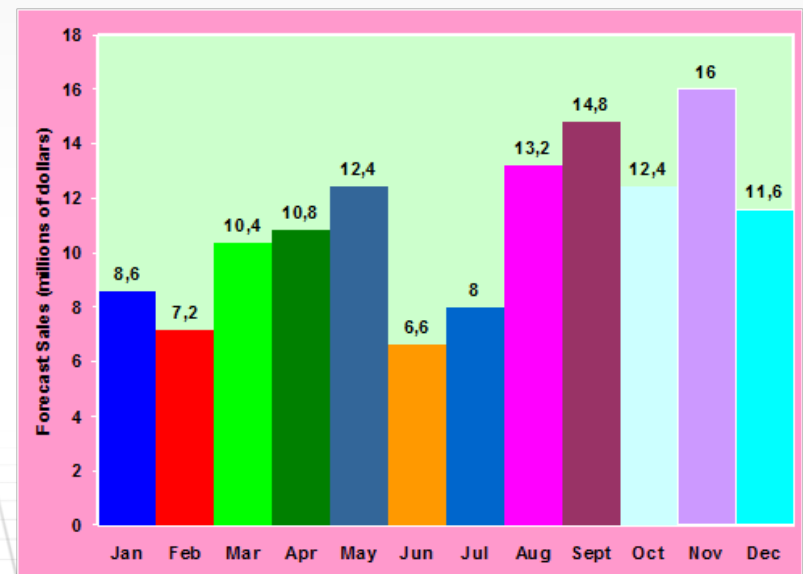
Capacity planning is concerned with determining the labor and equipment resources needed to achieve the master schedule.



Planning for Production (1)

Production planning is a process to develop tactical plans based on setting the overall level of manufacturing output (production plan) and other activities to best satisfy the current planned levels of sales, while meeting general business objectives of profitability, productivity, competitive customer lead times, and so on, as expressed in the overall business plan.

- ❑ The production plan states the aggregate production goals for all products manufactured by the enterprise.
- ❑ Converting the aggregate forecast into a production plan requires a decision on resources.





Planning for Production (2)

- ❑ In order to make a production plan three different production-planning strategies could be used:
 - Chase
 - Level
 - Mixed
- ❑ The chase production strategy requires that the production in each period equals the planned production for that period.
- ❑ A pure chase strategy requires that either the number of employees or the hours worked per week by each employee must change to meet the production planned for the period. (see the last two columns in Table 9.1)
- ❑ The level production strategy requires that the production in each period equals the monthly average production calculated from the total production value for the year. (see the Table 9.2)

Chase Production Strategy

Table 9.1 Chase Production Strategy Data

Month	Sales (mil. of \$) [S _i]	Labor (hr)	Days worked [d _i]	Variable workforce (number of employees required) [e _i]	Variable work week (hours per employee) [h _i]
Jan	8.6	86,000	21	512	29.72
Feb	7.2	72,000	20	450	26.12
Mar	10.4	104,000	23	565	32.81
Apr	10.8	108,000	20	675	39.19
May	12.4	124,000	22	705	40.90
Jun	6.6	66,000	10	825	47.89
Jul	8	80,000	21	476	27.64
Aug	13.2	132,000	22	750	43.54
Sept	14.8	148,000	21	881	51.14
Oct	12.4	124,000	22	705	40.90
Nov	16	160,000	20	1,000	58.05
Dec	11.6	116,000	20	725	42.09
Total	132	1,320,000	242		
Average			20	689	40

Sales dollar conversion estimated to be \$100 per employee hour

$$e_i = \frac{S_i}{s \times d_i \times h_d}$$

$$h_i = \frac{e_i \times H}{E}$$

S – sales in million of dollars

s – sales dollars estimated /employee/hr

d – days worked/month

e – number of employees required/month

E – average number of employees required/month

h – hours/employee/week

H – average number of hours/week

h_d – hours/day

Level Production Strategy

Table 9.2 Level Production Strategy Data

Month	Sales (mil. of \$) [S _i]	Labor (hr) [I _i]	Days worked [d _i]	Level workforce (number of employees required) [E]	Monthly production (millions of dollars) [mp _i]	Inventory balance (millions of dollars) [Ib _i]
Jan	8.6	86,000	21	689	11.58	2.98
Feb	7.2	72,000	20	689	11.02	6.80
Mar	10.4	104,000	23	689	12.68	9.08
Apr	10.8	108,000	20	689	11.02	9.30
May	12.4	124,000	22	689	12.13	9.03
Jun	6.6	66,000	10	689	5.51	7.94
Jul	8	80,000	21	689	11.58	11.51
Aug	13.2	132,000	22	689	12.13	10.44
Sept	14.8	148,000	21	689	11.58	7.22
Oct	12.4	124,000	22	689	12.13	6.94
Nov	16	160,000	20	689	11.02	1.97
Dec	11.6	116,000	20	689	11.02	1.39
Total	132	1,320,000				
Average			20	689	11.2	7.05

Planned work week is 40 hours long, Sales dollars estimated/employee/hr is \$100

$$I_i = \frac{S_i}{s}$$

$$L = \frac{\sum_{i=1}^m I_i}{m}$$

$$D = \frac{\sum_{i=1}^m d_i}{m}$$

$$E = \frac{L}{D \times h_d}$$

$$mp_i = E \times d_i \times h_d \times s$$

$$Ib_i = mp_i - S_i + Ib_{i-1}$$

$$i = 1 \div m, Ib_0 = 0$$

S - sales in million of dollars

s - sales dollars estimated /employee/hr

L - average number of labor hours/month

I - number of labor hours/month

m - number of months/year

D - average number of days/month

d - days worked/month

E - average number of employees required/month

h_d - hours/day

mp - monthly production

Ib - inventory balance

Mixed Production Strategy

Table 9.3 Mixed Production Strategy Data

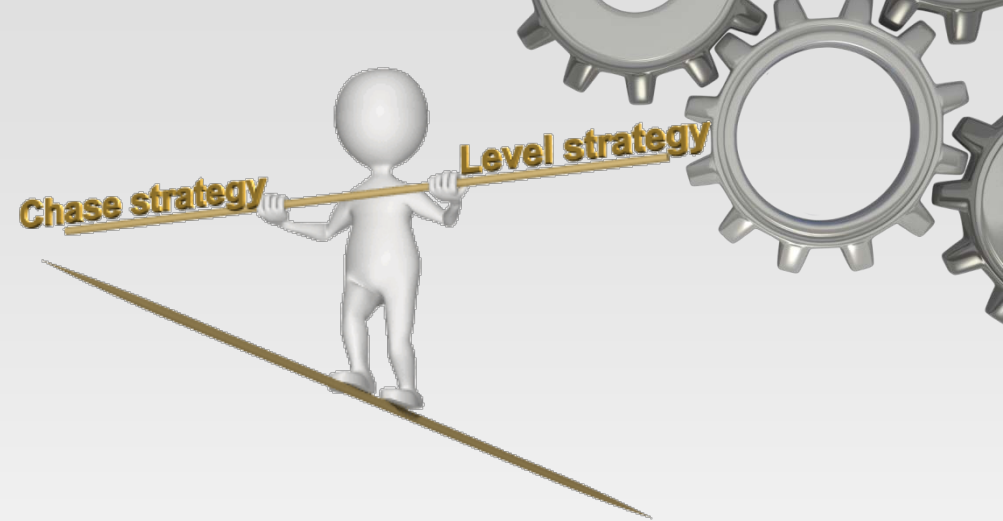
Month	Sales (mil. of \$)	Labor (hr)	Days worked	Level workforce (number of employees required)	Monthly production (millions of dollars)	Inventory balance (millions of dollars)
Jan	8.6	86,000	21	610	10.25	1.65
Feb	7.2	72,000	20	610	9.76	4.21
Mar	10.4	104,000	23	610	11.22	5.03
Apr	10.8	108,000	20	610	9.76	3.99
May	12.4	124,000	22	610	10.74	2.33
Jun	6.6	66,000	10	610	4.88	0.61
Jul	8	80,000	21	610	10.25	2.86
Aug	13.2	132,000	22	809	14.24	3.89
Sept	14.8	148,000	21	809	13.59	2.69
Oct	12.4	124,000	22	809	14.24	4.52
Nov	16	160,000	20	809	12.94	1.47
Dec	11.6	116,000	20	809	12.94	2.81
Total	132	1,320,000				
Average			20		11.23	3.00

Planned work week is 40 hours long, January starting inventory balance is zero

- With this method, the average inventory is reduced from \$7.05 million for the level strategy to \$3 million for the mixed approach.
- In most manufacturing situations, the mixed strategy is used to adjust the variable resources to meet the production-planning goals.



Chase versus Level Production Strategy



- ☐ The advantage of the chase strategy is that no excess inventory is carried.
- ☐ The disadvantage of the variable workforce implementation of chase is the high cost of hiring and firing employees.
- ☐ In the variable work week implementation of chase, the disadvantage is the cost of overtime and in implementing shortened work weeks.
- ☐ The advantage of the level strategy is the constant workforce and work week, but that constancy creates a disadvantage in the cost of carrying the excess inventory created.

Aggregate production planning and MPS (1)

- ❑ Aggregate production planning is a high-level corporate planning activity.
- ❑ The aggregate production plan indicates production output levels for the major product lines of the company.
- ❑ The production quantities of the major product lines listed in the aggregate plan (Table 9.4a) must be converted into a very specific schedule of individual products – master production schedule (MPS) (Table 9.4b)

Table 9.4 Aggregate production planning and MPS

Product line	Week									
	1	2	3	4	5	6	7	8	9	10
M model line	200	200	200	150	150	120	120	100	100	100
N model line	80	60	50	40	30	20	10			
P model line							70	130	25	100

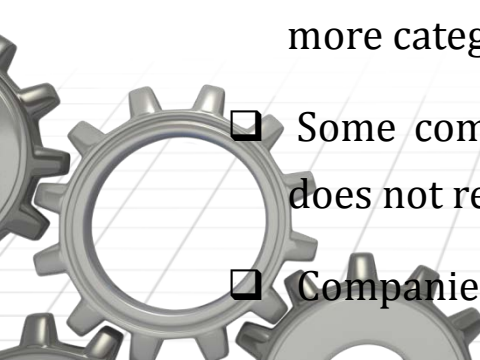
(a)

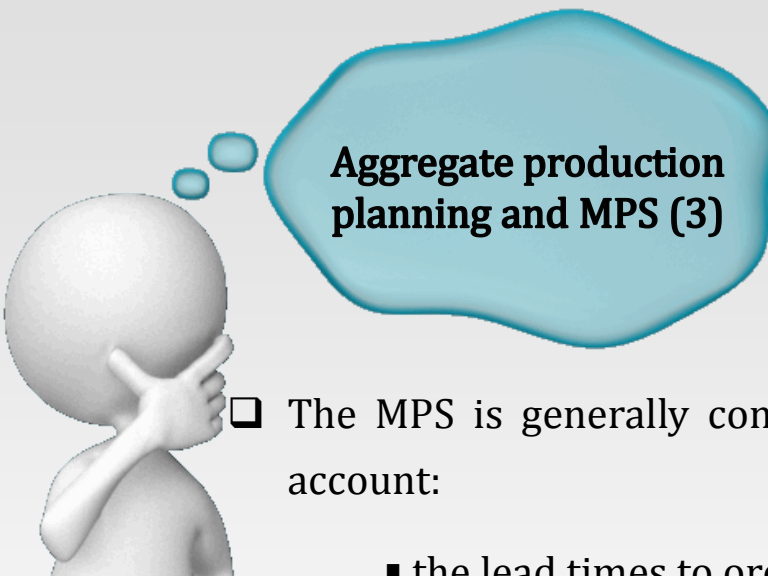
Product line models	Week									
	1	2	3	4	5	6	7	8	9	10
Model M3	120	120	120	100	100	80	80	70	70	70
Model M4	80	80	80	50	50	40	40	30	30	30
Model N8	80	60	50	40	30	20	10			
Model P1								50		100
Model P2							70	80	25	

(b)



Aggregate production planning and MPS (2)

- ❑ Products included in the MPS divide into three categories:
 - ***firm customers orders:*** the company is obligated to delivery the item by a particular date that has been promised by the sales department.
 - ***forecasted demand:*** production output quantities are based on statistical forecasting techniques applied to previous demand patterns, estimates by the sales staff and other sources.
 - ***spare parts:*** consists of repair parts that will either be stocked in the company's service department or sent directly to the customer.
 - ❑ Proportions in each category vary for different companies, and in some cases one ore more categories are omitted.
 - ❑ Some companies exclude the spare parts category from the master schedule since it does not represent products.
 - ❑ Companies producing assembled products will generally have to handle all three types.
- 



Aggregate production planning and MPS (3)

- ❑ The MPS is generally considered to be a medium-range plan since it must take into account:
 - the lead times to order raw materials and components
 - produce parts in the factory
 - assemble the end products.
- ❑ Lead times can range from several weeks to many months (even years).
- ❑ MPS is usually considered to be fixed in the near term.
- ❑ Other inputs that may cause the master schedule to depart from the aggregate plan include:
 - new customer orders
 - changes in sales forecast over the near term

Aggregate production planning and MPS (4)

Table 9.5 MPS Time-Phased Record

	Period									
	1	2	3	4	5	6	7	8	9	10
Forecast	5	5	5	5	5	5	20	20	20	20
Available	26	32	38	44	50	56	47	38	29	20
MPS	11	11	11	11	11	11	11	11	11	11
On hand	20									

Table 9.6 MPS Time-Phased Record with lot size

	Period									
	1	2	3	4	5	6	7	8	9	10
Forecast	5	5	5	5	5	5	20	20	20	20
Available	45	40	35	60	55	50	60	40	20	0
MPS	30			30			30			
On hand	20									

$$MPS_i = \frac{\sum_{i=1}^{No.of\ periods} Forecast_i}{No.of\ periods}$$

$$Available_1 = OnHand + MPS_1 - Forecast_1$$

$$Available_i = Available_{i-1} + MPS_i - Forecast_i \quad i = 2..NoOfPeriods$$

- The number of periods indicated on the record is a function of the individual industry and product.
- The forecast row represents the number of end-item units that the business anticipates will be sold in each period.
- The available row represents the inventory balance at the end of the period or the number of units available for sale in the next period.
- The MPS row indicates the number of units scheduled for production during the period and available to satisfy the forecast for the period.
- The On Hand value is the number of units present in inventory at the start of the first period.

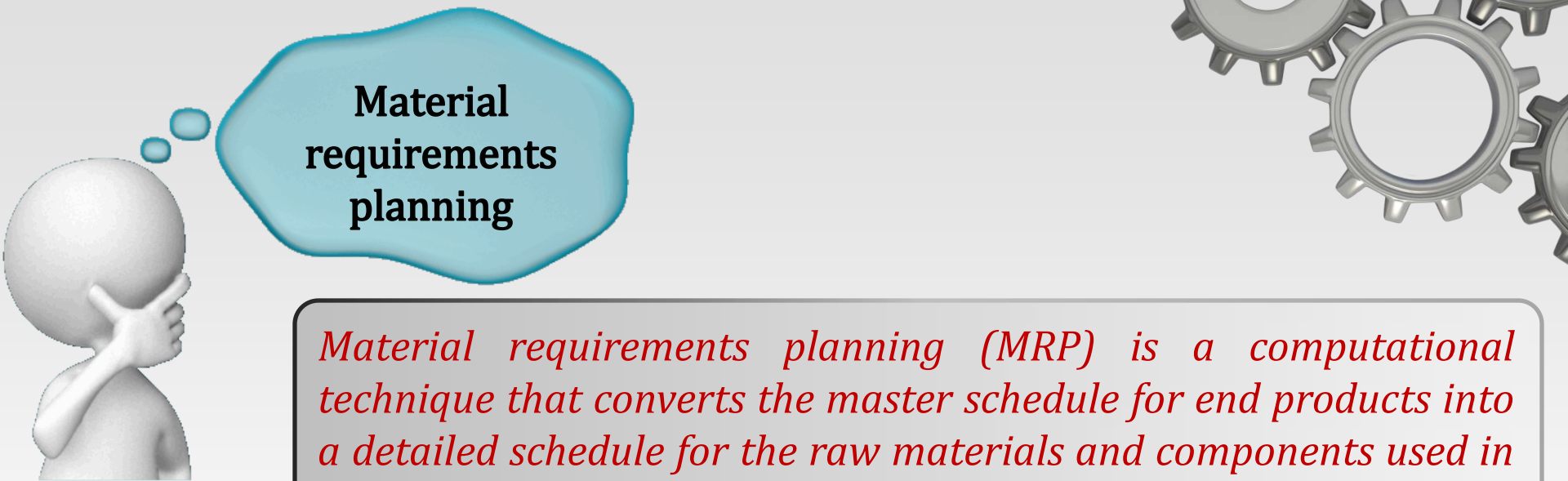
Aggregate production planning and MPS (5)

- ❑ The time-phased records illustrated in Table 9.5 and 9.6 do not allow for production situations where customers place orders for future delivery.
- ❑ In reality sales planned for future periods may occur as an order from a customer in an earlier period.
- ❑ One way to handle this additional variation in planning the master schedule is to expand the time-phased record to include two more rows ATP (Available To Promise) and Orders.

Table 9.7 MPS Time-Phased Record with Order Promising

	Period									
	1	2	3	4	5	6	7	8	9	10
Forecast	10	10	10	10	10	15	15	15	15	15
Orders	6	5	3							
Available	30	20	10	30	20	5	20	5	20	5
ATP	26			30			30		30	
MPS	30			30			30		30	
On hand	10									

- The Orders row indicates all orders received for future products that will be consuming the forecast.
- The quantity listed in the Forecast row is expected to be sold and the values in the Order row represent firm sales of the product.
- The ATP value for period 1 is calculated by adding the On Hand and MPS values and subtracting the total orders received up to period 3.



Material requirements planning

Material requirements planning (MRP) is a computational technique that converts the master schedule for end products into a detailed schedule for the raw materials and components used in the end products.

❑ The detailed schedule :

- identifies the quantities of each raw material and component item;
 - indicates when each item must be ordered and delivered to meet the master schedule for final products.
-
- *independent demand* -> demand for a product is unrelated to demand for other items
 - *dependent demand* -> demand for the item is directly related to the demand for some other items, usually a final product.





Self-assessment test

1. The constant production more often creates large stocks.

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2. In the case of the variable production, labor demand remains constant if you can vary the number of hours worked.

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3. Master production schedule is generally developed in the medium term.

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4. Independent demand is the one in which demand for a product does not depend on the demand for another product.

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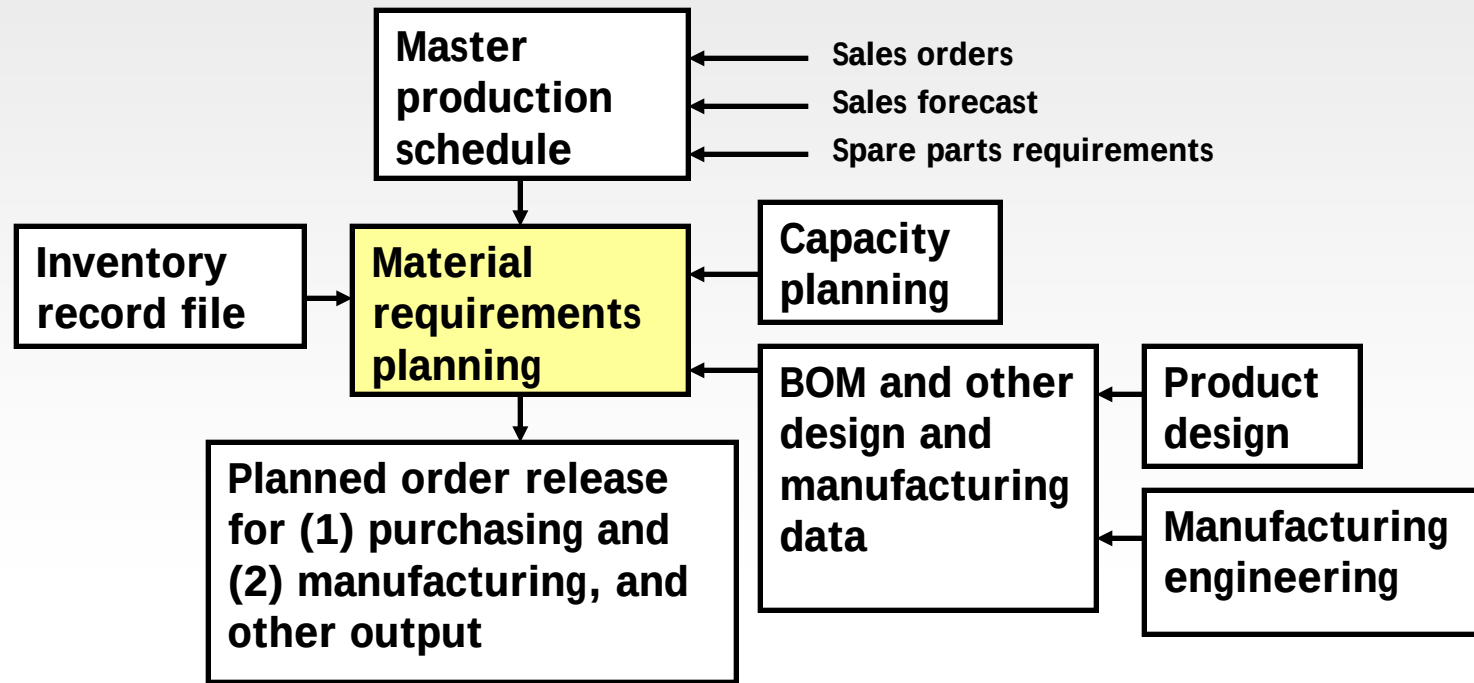
5. Master production schedule must consider the information related to capabilities.

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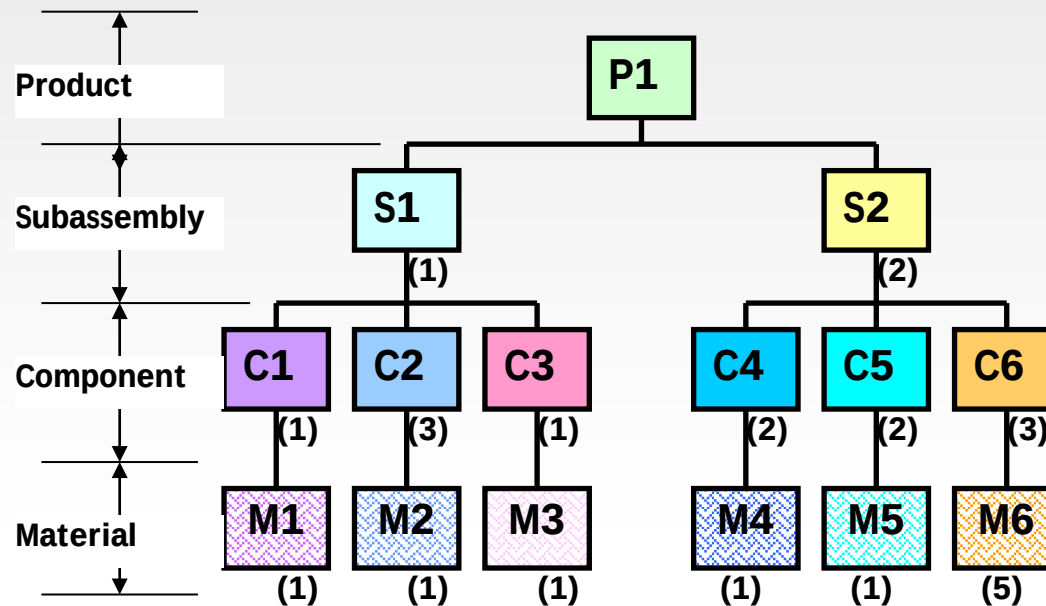
Inputs to the MRP system



A possible structure of a product

Example 1

The figure below shows the possible structure of a product (P1), under which it can establish the necessary materials:



- The number between brackets represent the amount of that subassembly, that component or that material.
- The time to produce each component shall be taken into account.

The MRP record (1)

- ❑ The record shown in Table 9.8 represents the production plans for the part number under study.
- ❑ The accepted convention considers period 1 the current period and periods 2 through 8 to be in the future.

Table 9.8 MRP record for part #2656

Item: #2656 brass ring	Period							
	1	2	3	4	5	6	7	8
Gross requirements				100				
Scheduled receipts								
Projected on hand	140	140	140	40	40	40	40	40
Planned order receipts								
Planned order releases								
Order Policy = Fixed lot Order Quantity = 200 Lead time = 5								

- ❑ **Gross requirements** equal the anticipated future demand, both independent and dependent, for an item, stated period by period, not aggregated or averaged.
- ❑ **Scheduled receipts** represent all orders released to manufacturing or to suppliers through purchase orders.
- ❑ **Projected on hand** represents the calculated inventory for the item projected through all the periods on the record. The projected available balance is the running sum of on-hand inventory minus gross requirements plus scheduled receipts and future planned order releases. The value for the projected available balance represents the inventory at the end of the period in which it appears.
- ❑ **Planned order receipts** indicates when a planned order will be received if the planned order release date is exercised.
- ❑ **Planned order releases** are the suggested order quantity, release date, and due date generated with MRP software.

The MRP record (2)

- The gross requirements (16 units) (Table 9.9) come from either the next-highest level in the product structure diagram or from the MPS.
- The scheduled receipts value represents planned order releases that became firm orders when the orders were placed.
- The on-hand inventory for the start of the first period (4 units listed in the box) is determined from the MRP record from the last period.
- The lead time, lot size, and safety stock are values set by purchasing and manufacturing.
- The calculations for the record start with the first period and proceed to the last.
- If the projected inventory balance is negative or less than the required safety stock, a planned order release is required

Table 9.9 MRP record for part #2985- initial values

Item: #2985 silver ring	Period									
	1	2	3	4	5	6	7	8	9	10
Gross requirements	16		8	15	21		12	15		28
Scheduled receipts	15									
Projected on hand	4									
Planned order receipts										
Planned order releases										
Lead time = 1										
Lot size = 5										
Safety stock = 0										

Table 9.10 MRP record for part #2985- completed

Item: #2985 silver ring	Period									
	1	2	3	4	5	6	7	8	9	10
Gross requirements	16		8	15	21		12	15		28
Scheduled receipts	15									
Projected on hand	4									
Planned order receipts										
Planned order releases										
Lead time = 1										
Lot size = 5										
Safety stock = 0										

Supplies gross requirements for next record in product structure

The MRP record calculations

Period 1 calculations:

$$\begin{array}{rclclcl} \text{Starting} & & \text{Scheduled} & & \text{Gross} & & \text{Projected} \\ \text{Inventory} & + & \text{receipts} & - & \text{requirements} & = & \text{available} \\ 4 & & 15 & & 16 & & \text{balance} \\ & & & & & & 3 \end{array}$$

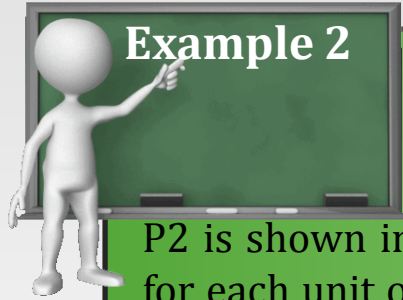
Period 2 calculations:

$$\begin{array}{rclclcl} \text{Starting} & & \text{Scheduled receipts} & & & & \text{Projected available} \\ \text{Inventory} & + & \text{(or planned order} & - & \text{Gross requirements} & = & \text{balance} \\ 3 & & \text{receipts)} & & 0 & & 3 \\ & & 0 & & & & \end{array}$$

- The starting inventory for the period 2 is the ending inventory for period 1.
- Period 2 could have either a scheduled receipt or a planned order receipt, depending on the production needs and the lead time.
- The equations for period 3 and for all subsequent periods will be the same as the equation given for period 2, and the results are shown in Table 9.10

Item: #2985 silver ring	Period									
	1	2	3	4	5	6	7	8	9	10
Gross requirements	16		8	15	21		12	15		28
Scheduled receipts	15									
Projected on hand	4	3	3	0	0	4	4	2	2	4
Planned order receipts			5	15	25		10	15		30
Planned order releases		5	15	25		10	15		30	
Lead time = 1	Supplies gross requirements for next record in product structure									
Lot size = 5										
Safety stock = 0										

MRP record



Example 2

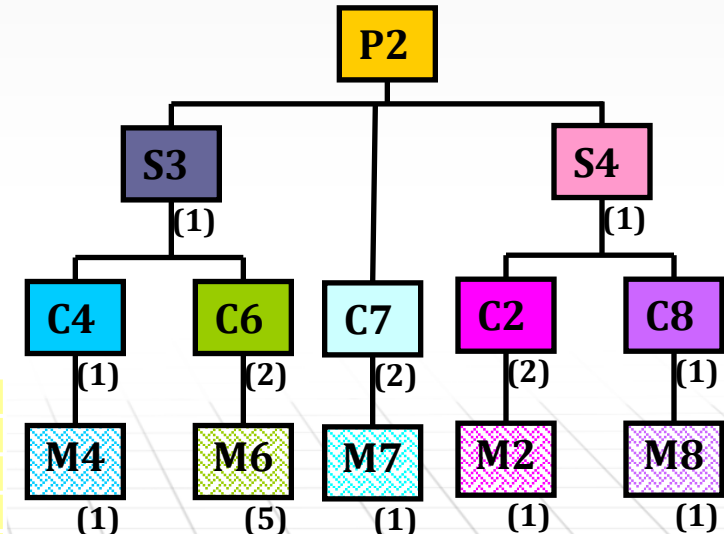
To illustrate various complicating factors, consider the MRP procedure for component C4, which is used in product P1. This part also happens to be used on product P2 of the master schedule in Table 9.4. The product structure for P2 is shown in the figure below. Component C4 is made out of material M4, one unit of M4 for each unit of C4, and the inventory status of M4 is given in Table 9.11. The lead times and inventory status for each of the other items needed in the MRP calculations are shown in the Table 9.12. Complete the MRP calculations to determine the time-phased requirements for items S2,S3,C4, and M4, based on the requirements for P1 and P2 given in the MPS of Table 9.4. We assume that the inventory on hand or on order for P1, P2, S2, S3, and C4 is zero for all future periods except for the calculated values in this problem solution.

Table 9.11 Initial inventory status of material M4

Period	1	2	3	4	5	6	7
Item: Raw material M4							
Gross requirements							
Scheduled receipts			40				
On hand	50	50	90				
Net requirements							
Planned order releases							

Table 9.12

Item	Lead time	Inventory
P1	Assembly lead time=1 wk	No inventory on hand or on order
P2	Assembly lead time=1 wk	No inventory on hand or on order
S2	Assembly lead time=1 wk	No inventory on hand or on order
S3	Assembly lead time=1 wk	No inventory on hand or on order
C4	Manufacturing lead time=2 wk	No inventory on hand or on order
M4	Ordering lead time=3 wk	



MRP record

Example 2 Solution

The results of the MRP calculations are given in Table 9.13.

The delivery requirements for P1 and P2 must be offset by their 1 wk assembly lead time to obtain the planned order release.

These quantities are then exploded into requirements for subassemblies S2(for P1) and S3(for P2). These requirements are offset by their 1 wk assembly lead time and combined in week 6 to obtain gross requirements for component C4.

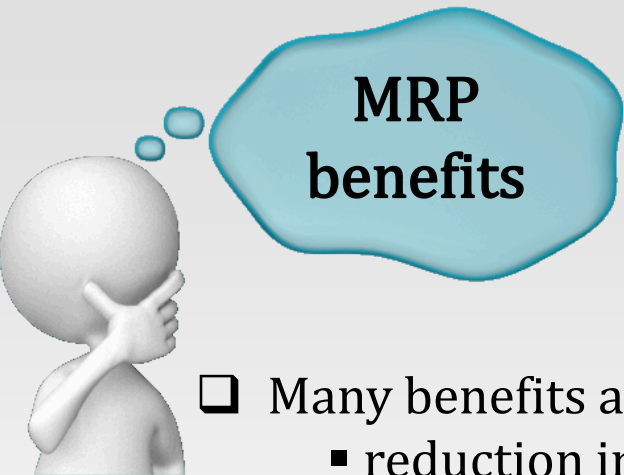
Net requirements equal gross requirements for P1,P2,S2,S3, and C4 because of no inventory on hand and no planned orders.

The on-hand stock of 50 units plus scheduled receipts of 40 are used to meet gross requirements of 70 units of M4 in week 3, with 20 units remaining that can be applied to the gross requirements of 280 units in week 4.

Net requirements in week 4 will be 260 units. With an ordering lead time of 3 weeks, the order release for 260 units must be planned for week 1.

Table 9.13

Period	1	2	3	4	5	6	7	8	9	10
Item: Product P1										
Gross requirements								50		100
Scheduled receipts										
On hand	0									
Net requirements								50		100
Planned order releases							50		100	
Item: Product P2										
Gross requirements							70	80	25	
Scheduled receipts										
On hand	0									
Net requirements							70	80	25	
Planned order releases						70	80	25		
Item: Subassembly S2										
Gross requirements							100		200	
Scheduled receipts										
On hand	0									
Net requirements							100		200	
Planned order releases						100		200		
Item: Subassembly S3										
Gross requirements						70	80	25		
Scheduled receipts										
On hand	0									
Net requirements						70	80	25		
Planned order releases					70	80	25			
Item: Component C4										
Gross requirements					70	280	25	400		
Scheduled receipts										
On hand	0									
Net requirements					70	280	25	400		
Planned order releases			70	280	25	400				
Item: Raw material M4										
Gross requirements			70	280	25	400				
Scheduled receipts			40							
On hand	50	50	90	20						
Net requirements			-20	260	25	400				
Planned order releases	260	25	400							



MRP benefits

- ❑ Many benefits are claimed for a well-designed MRP system:
 - reduction in inventory;
 - quicker response to changes in demand that is possible with a manual requirements planning system;
 - reduced setup and product changeover cost;
 - better machine utilization;
 - improved capacity to respond to changes in the master schedule;
 - as an aid in developing the master schedule

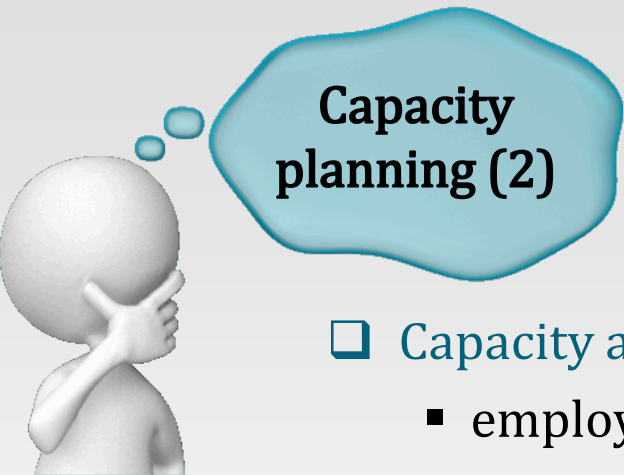
- ❑ Reasons why some MRP systems have not been successful include:
 - the application was not appropriate;
 - the MRP computations were based on inaccurate data;
 - the MPS was not coupled with a capacity planning system.



Capacity planning (1)

Capacity planning is concerned with determining what labor and equipment resources are required to meet the current MPS as well as long term future production needs of the firm.

- ❑ Capacity planning is typically accomplished in two stages:
 - **MPS stage:** a rough-cut capacity planning calculation is made to assess the feasibility of the master schedule,
 - **MRP stage:** a capacity requirements planning calculation determines whether there is sufficient production capacity in the individual departments.
- ❑ Capacity adjustments can be divided into:
 - short term adjustments,
 - long term adjustments.



Capacity planning (2)

□ Capacity adjustments for the short term include:

- employment levels,
- temporary workers,
- number of work shifts,
- labor hours,
- inventory stockpiling,
- order backlogs,
- subcontracting.

□ Capacity adjustments for the long term include following types of decisions:

- new equipment investments,
- new plant construction,
- acquisition of existing companies,
- plant closings.



Self-assessment test

1. Capacity planning deals with the allocation of equipment.

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2. Production planning involves planning labor and equipment required.

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3. Aggregate production planning deals with the allocation of human resources.

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4. Making a production plan involves the exclusive use of a production strategy according with the demand.

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5. In the case of level production strategy, the number of employees and the hours worked are variable.

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Conclusion



- ☐ Production planning is a particularly important activity because it determines the market competitiveness of producers.
- ☐ A good materials requirements planning will reduce the inventory and the delivery time.
- ☐ Adjustment of production capacities is an activity that involves decisions on both short and long term.





- ☐ Shop floor control
- ☐ Inventory control
- ☐ Manufacturing resource planning (MRP II)
- ☐ Just-in-time production systems